Module 05: Action in Crochet Circles

Pulling Through, Letting Go

Learning Objectives

1. Describe tension adjustment as a form of active inference — the crocheter continuously corrects to maintain desired fabric properties.
2. Explain yarn feeding technique and switching between yarns as action policies that minimize prediction error.
3. Connect the physical act of pulling through a loop to the action component of the perception-action cycle.

Introduction

Every crochet stitch is an action. The hook enters the fabric, catches the yarn, and pulls through — a sequence so fundamental that experienced crafters perform it thousands of times in a single project without conscious thought. But each of those actions is a micro-decision, shaped by what the hands expect and what they find.

In Active Inference, **action** is how an agent changes the world to bring it into line with its predictions. The crocheter does not passively accept whatever the yarn does. They actively shape the yarn — adjusting tension, controlling feed rate, choosing hook angle — to produce the fabric they expect.

This module explores the action side of the crocheter-yarn relationship: how tension works as continuous active inference, how yarn feeding is a trained action policy, and how the simple act of pulling through a loop embodies the entire perception-action cycle.

Key Concepts

1. Tension Adjustment as Active Inference

Tension is the crocheter's primary action parameter. It is the pressure with which the hands hold and feed the yarn, and it determines nearly everything about the resulting fabric — gauge, drape, stiffness, stitch definition.

Tension is not set once and forgotten. It is continuously adjusted, stitch by stitch, based on the difference between what the crocheter expects the fabric to feel like and what it actually feels like. If the fabric feels too tight, the hands ease up. If it feels too loose, the hands pull firmer. This is the perception-action loop operating at its most basic level.

Different yarns require different tension. A slippery silk needs to be held more firmly to prevent the stitches from loosening as you work. A grippy cotton holds its shape well and can be worked with lighter tension. A springy wool bounces back against the hook, requiring a particular negotiation between hand pressure and fiber elasticity.

2. Yarn Feeding as Action Policy

How the yarn travels from the skein to the hook is not random — it follows a specific path through the crocheter's fingers. This path is an **action policy**: a learned, optimized sequence of finger positions that controls the yarn's speed, tension, and alignment.

Common feeding methods include: - **Wrapped around the pinky**: The yarn wraps once around the little finger, then travels across the palm and over the index finger. The pinky acts as a brake, controlling feed rate. - **Threaded between fingers**: The yarn weaves between the fingers in a figure-eight pattern, creating multiple friction points for fine tension control. - **Tensioned over the index**: The yarn runs over the index finger and is controlled by varying the finger's height and curl.

Each method is a different action policy, optimized for different hands, different yarns, and different stitch patterns. Crafters often discover their preferred method through trial and error — which is itself a form of active inference, testing different policies and settling on the one that minimizes prediction error (produces the most consistent, comfortable results).

3. The Pull-Through: A Micro-Action Cycle

The fundamental crochet action — insert hook, yarn over, pull through — is a complete perception-action cycle compressed into a fraction of a second:

1. **Predict**: The hands expect the hook to enter at a certain point with a certain resistance.
2. **Act**: The hook is inserted. If the resistance matches prediction, the action continues. If not (the stitch is too tight, the hook catches a split ply), the hands adjust.
3. **Predict**: The hands expect the yarn to catch on the hook at a certain angle.
4. **Act**: The yarn is wrapped over the hook. Adjustments are made for yarn slipperiness, hook throat shape, and stitch type.
5. **Predict**: The hands expect a certain resistance as the loop is pulled through.
6. **Act**: The loop is pulled through. The resulting stitch is immediately evaluated — does it look and feel right?

This six-step cycle happens in less than a second for experienced crocheters. It is entirely embodied — no conscious deliberation required. The hands think through action.

4. Switching Between Yarns

When a project requires multiple yarns — different colors in a stripe pattern, different fibers for contrast — the crocheter must switch between action policies. Each yarn has its own optimal tension, feed rate, and hook behavior.

This switch is a moment of heightened prediction error. The hands were calibrated for Yarn A, and now they must rapidly recalibrate for Yarn B. Experienced crafters handle this smoothly because their generative model includes multiple yarn-specific action policies that can be swapped in and out. Beginners may struggle because they have only one policy (or none that is well-trained).

5. Letting Go: The Active Inference of Release

Not all action in crochet is about adding — some is about releasing. Letting the yarn feed freely through the fingers after a stitch, allowing the fabric to relax and settle, even the decision to frog — these are all active choices shaped by prediction.

Letting go is particularly important in: - **Blocking**: Releasing the pinned fabric to see how it settles - **Tension correction**: Consciously relaxing the grip to ease overly tight work - **Yarn joins**: Releasing the old yarn and picking up the new one - **Finishing**: Cutting the yarn and pulling through the final loop

Each of these release actions is driven by the same active inference process — the agent predicts what will happen when it lets go, and the result either confirms or contradicts that prediction.

The Free Energy Principle and Action

The Free Energy Principle says that agents act to minimize surprise — to make the world conform to their predictions. In crochet, this means the hands are constantly working to produce fabric that matches the crocheter's expectations. When it does, free energy is low and the action feels smooth. When it does not, free energy rises and the hands must work harder.

This is why skilled crochet looks effortless — the actions are so well-calibrated that the fabric seems to form itself. The effort is invisible precisely because the prediction errors are tiny and the corrections are instantaneous.

Conclusion

Every stitch is an action, and every action is an inference. Tension adjustment, yarn feeding, the pull-through, yarn switching, and even letting go — all are driven by the crocheter's predictions about what the yarn should do and the hands' continuous effort to make it so. In the next module, we turn to learning — how the crocheter discovers what each yarn can do through experience.

Connections

- **Previous Module**: Cognition — When Your Hands Think Ahead
- **Next Module**: Learning — Discovering What Each Yarn Can Do
- **Course Home**: Fiber & Flow